

Qatar-1b

R-0413

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_170619 · created 2026-07-19T06:38:49+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457923.9331 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.055
Transit depth (Rp/Rs)^2	3.67 +/- 2.09 [%]
Orbital Inclination (inc)	82.63 +/- 2.8
Airmass coefficient 1 (a1)	42.43 +/- 26.35
Airmass coefficient 2 (a2)	-2.99 +/- 1.68
Scatter in the residuals of the lightcurve fit is	62.32 %
Best Comparison Star	None
Optimal Aperture	0.75
Optimal Annulus	5.75
Transit Duration (day)	0.056 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.192 ± 0.055 vs 0.1463 (deviation 0.78σ)
Duration fit vs published	0.056 d vs 0.0692 d (deviation 0.51σ)
Mid-time O–C	1.3 min
Depth SNR	3.3
Residual scatter	62.32 %

Light curve

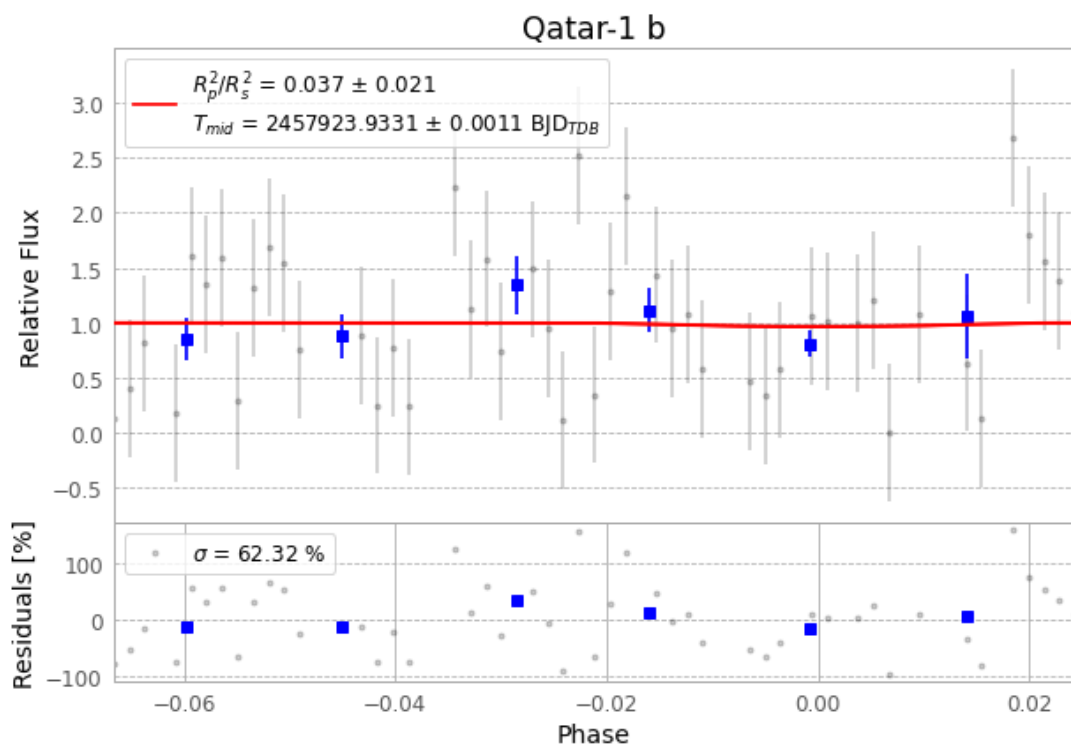


Figure 1 — FinalLightCurve_Qatar-1 b_2017-06-19.png (SHA-256 76e2aa2318ae9439...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [398, 180], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7ccbe75a49f4f046...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

64 FITS frames (42 MB), Qatar-1170619080612.FITS → Qatar-1170619111512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-170619.log (57de5869b821...), FinalLightCurve_Qatar-1 b_2017-06-19.png (76e2aa2318ae...), FinalLightCurve_Qatar-1 b_2017-06-19.csv (96a553c04b3f...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 06:38Z.